

Time-Varying Volatility versus Heavy Tails in S&P 500 Returns: a Bayesian Analysis

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Abstract

Stock returns are heavy-tailed, and calm and turbulent periods tend to cluster in time (volatility clustering). First, we ask whether the heavy tails are simply a consequence of volatility changing over time, and, second, whether splitting the market into “normal” and “crisis” regimes helps to forecast volatility. Using Bayesian stochastic volatility models on weekly returns from 2005 to 2025, we find that volatility changes strongly over time: shocks fade with a half-life of about 14 weeks. Once this is taken into account, the remaining randomness is close to Normal: the heavy tails come mainly from changing volatility rather than from extreme individual shocks. On daily data, a model with a normal and a crisis regime for log realized volatility is evaluated out of sample (2023–2025). It performs the same as a single-regime AR(1) and only marginally better than the naive persistence forecast, and its crisis signal is lagged by construction. Time-varying volatility is the key feature of the data, while discrete regimes add no predictive value in this setting.

1 Introduction

Two well-known facts of asset returns are heavy tails and volatility clustering: returns are nearly uncorrelated, but their absolute values are strongly and persistently autocorrelated [1]. These two facts are related. If returns are conditionally Normal but their variance changes over time, their unconditional distribution is a scale mixture of Normals, which is heavy-tailed even though each individual return is not. This raises a modelling question: once volatility is allowed to vary over time, are return shocks still heavy-tailed?

We study this question in a fully Bayesian way, with two contributions:

1. We fit **stochastic volatility** models [2, 3] with Student- t innovations [4] to weekly S&P 500 returns. We use the posterior of the degrees of freedom ν to quantify how heavy-tailed the return shocks remain once volatility varies over time.
2. We test whether a **threshold** (“crisis regime”) **model** for daily realized volatility [5] improves one-step-ahead predictions over simple benchmarks on a held-out period.

All models are implemented with the PyMC Python library and their posteriors are sampled with NUTS [6]. Convergence of every parameter and latent state is verified with the diagnostics recommended by Vehtari et al. [7]: $\hat{R}$, effective sample size and divergent transitions.

2 Data

We use daily closing prices of the S&P 500 index (`^GSPC`, Yahoo Finance) from 3 January 2005 to 31 December 2025. Weekly prices are taken as the last close of each week. The data is kept

*Revised version of a group project for the *AI Programming* course at TU Wien, originally developed with M. Csikós. Code and data: <https://github.com/niccolo-s/bayesian-volatility-regimes>.

as a frozen snapshot, so that every result can be reproduced. Prices are converted to percentage log-returns, $r_t = 100 (\ln P_t - \ln P_{t-1})$, the standard scale in the SV literature [3].

Table 1 confirms the two facts discussed above. Excess kurtosis is large at both frequencies, the autocorrelation of $|r_t|$ is about 0.3, while that of r_t is weaker and slightly negative.

	n	sd (%)	skewness	excess kurtosis	ACF(1) of r_t	ACF(1) of $ r_t $
Daily	5282	1.21	−0.48	13.2	−0.12	0.32
Weekly	1095	2.45	−0.95	8.4	−0.08	0.34

Table 1: Descriptive statistics of S&P 500 percentage log-returns, 2005–2025.

3 Stochastic volatility on weekly returns

3.1 Models

Both models share the equation

$$r_t = e^{h_t/2} \varepsilon_t, \quad \varepsilon_t \sim t_\nu(0, 1), \quad (1)$$

where h_t is the latent log-variance. They differ in the model for h_t :

- **SV-RW:** a Gaussian random walk, $h_t = h_{t-1} + \sigma_h \eta_t$;
- **SV-AR(1):** a stationary AR(1), $h_t = \mu + \phi(h_{t-1} - \mu) + \sigma_\eta \eta_t$, with $h_1 \sim \mathcal{N}\left(\mu, \frac{\sigma_\eta^2}{1-\phi^2}\right)$

where $\eta_t \sim \mathcal{N}(0, 1)$.¹

Table 2 lists the priors. For the AR(1) model we follow Kim et al. [3, Sec. 2.2.1]. For the random walk we use weakly informative, plausible priors [8, Sec. 2.9]; for ν we use the Gamma(2, 0.1) prior of Juárez and Steel [9].

Model	Parameter	Prior	Source
SV-AR(1)	μ	$\mathcal{N}(0, 10)$	Kim et al. [3]
	$(\phi + 1)/2$	Beta(20, 1.5)	Kim et al. [3]
	σ_η^2	Inv-Gamma(2.5, 0.025)	Kim et al. [3]
SV-RW	σ_h	Exp(5)	weakly informative
	h_1	$\mathcal{N}(\log \hat{\sigma}, 0.5^2)$	weakly informative
Both	ν	Gamma(2, 0.1)	Juárez and Steel [9]

Table 2: Prior distributions. $\hat{\sigma}$ is the sample standard deviation of weekly returns.

The latent path is written in non-centered form: we sample standardized shocks η_t and build h_t from them. This avoids the strong dependence between the volatility-of-volatility parameter (σ_h and σ_η) and h_t that otherwise makes Hamiltonian Monte Carlo unstable [10].

3.2 Results

Both models converge without divergent transitions ($\hat{R} < 1.005$ for all parameters and all 1095 latent states). Table 3 reports the statistics of the posteriors.

¹In the SV-RW model h_t is the log standard deviation ($r_t = e^{h_t} \varepsilon_t$) instead of the log-variance: the two differ only by a factor of 2.

Model	Parameter	Posterior mean	94% HDI
SV-RW	σ_h	0.113	[0.087, 0.142]
	ν	31.0	[10.5, 56.9]
SV-AR(1)	μ	1.20	[0.85, 1.56]
	ϕ	0.948	[0.920, 0.975]
	σ_η	0.282	[0.209, 0.350]
	ν	35.2	[11.4, 62.0]

Table 3: Posterior summaries of the SV models on weekly returns (4 chains $\times$ 1000 draws).

Volatility dynamics. Volatility changes strongly over time (Figure 1). Its peaks coincide with well-known episodes: the 2008–09 financial crisis, the 2011 sell-off, the COVID-19 crash of March 2020, the 2022 bear market and the April 2025 sell-off. Volatility shocks are persistent ($\phi = 0.948$) and halve in about 14 weeks. The typical volatility level, $e^{\mu/2} \approx 1.8\%$ per week, corresponds to about 13% per year.

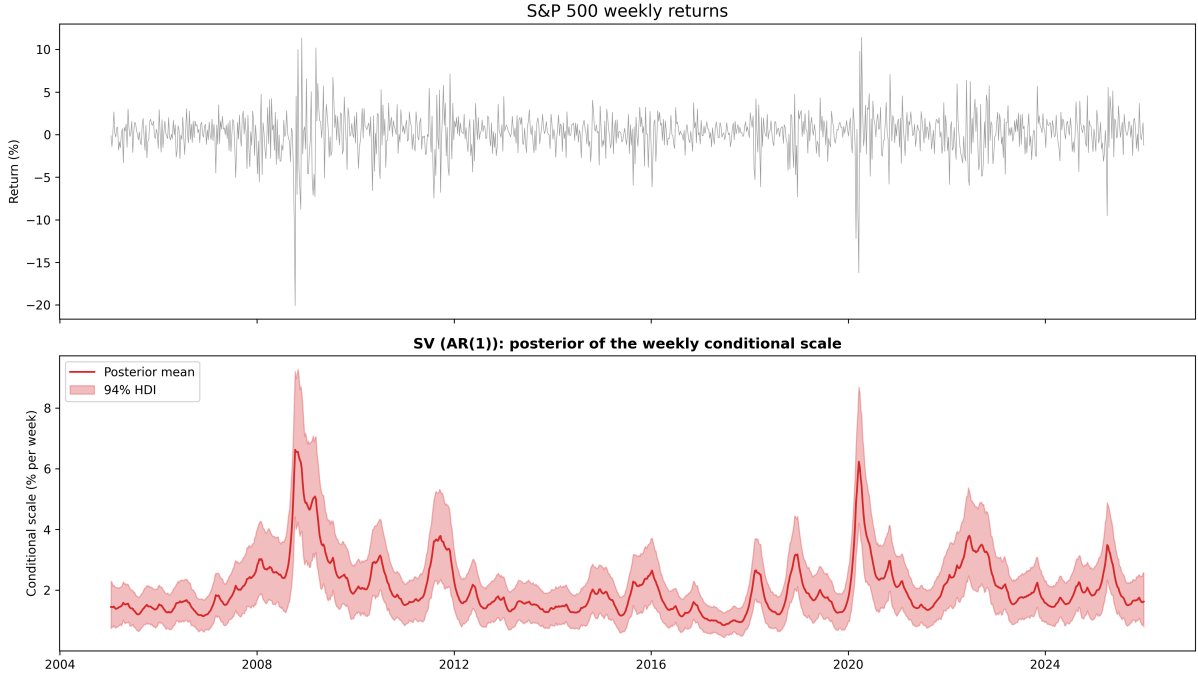


Figure 1: Top: weekly S&P 500 returns. Bottom: estimated weekly volatility $e^{h_t/2}$ from the SV-AR(1) model, with posterior mean (line) and 94% credible interval (band).

How heavy are the tails once volatility varies? The data clearly rule out small values of ν : the prior puts 26% of its mass below 10, while both posteriors’ credibility intervals lie above $\nu \approx 10.5$. Large values cannot be distinguished, so the upper end of the posterior reflects the prior. A Student- t with $\nu \geq 10.5$ has an excess kurtosis of at most $6/(\nu - 4) \approx 0.9$, far below the 8.4 of weekly returns. The return shocks are therefore close to Normal, and the heavy tails come mainly from changing volatility, in line with Jacquier et al. [4] for daily data.

Robustness to the volatility dynamics. We take the stationary AR(1) as the main configuration, as is standard in the SV literature [3]; the random walk serves as a robustness check. Both models give the same qualitative picture: a strongly time-varying volatility and return shocks with the ν parameter well above 10. The conclusions on the tails therefore hold regardless of how the dynamics of h_t are specified.

4 Do volatility regimes help forecasting?

4.1 Threshold model and benchmarks

An alternative to a continuously varying volatility is a model with two discrete regimes, “normal” and “crisis”. We model the daily log realized volatility $h_t = \log RV_t - \overline{\log RV}_{\text{train}}$, where RV_t is the standard deviation of the last five daily returns and $\overline{\log RV}_{\text{train}}$ is its mean over the training period, with a threshold AR(1) [5]:

$$h_t = \mu_{S_t} + \phi_{S_t} h_{t-1} + \sigma_{S_t} \epsilon_t, \quad \epsilon_t \sim t_\nu(0, 1), \quad S_t = \mathbb{1}_{\{RV_{t-1} > \gamma\}} \quad (2)$$

Day t is in the crisis regime ($S_t = 1$) if the previous day’s realized volatility exceeds γ , the 90th percentile of the training period (1.78% per day). The regime therefore uses only past information, both in estimation and in forecasting. The priors favour persistent volatility ($\phi_S \sim \text{Beta}(20, 2)$ in the normal regime, $\text{Beta}(15, 3)$ in the crisis regime, both with most of their mass close to 1), and $\nu \sim \text{Gamma}(2, 0.1)$, as before. With over 4500 training days, the posteriors are dominated by the data.

We compare this model with two benchmarks: a **single-regime AR(1)**, identical but without regimes, which shows whether the regimes add any information; and the **persistence** forecast $\hat{h}_t = h_{t-1}$, i.e. “tomorrow equals today”. All parameters are estimated on 2005–2022 (4525 days) and kept fixed over the 2023–2025 test period (752 days). For each test day we perform the one step ahead prediction of h_t and measure: the mean absolute error of the forecast, the share of observations falling inside the 90% predictive interval, and the mean log predictive density, a standard score for probabilistic forecasts [11].

4.2 Results

Table 4 and Figure 2 summarize the results. The threshold model is practically indistinguishable from the single-regime AR(1): the MAEs and the log scores are comparable. Both are only marginally better than persistence. Their predictive intervals are well calibrated (coverage 0.887 against a nominal 0.90).

Model	MAE	90% coverage	mean log pred. density
Threshold AR(1)	0.1941	0.887	−0.0644
Single-regime AR(1)	0.1943	0.887	−0.0641
Persistence $\hat{h}_t = h_{t-1}$	0.1966	—	—

Table 4: Out-of-sample one-step-ahead forecasts of centered log realized volatility, 2023–2025.

Three factors likely explain this outcome.

- **Overlapping windows.** Consecutive five-day windows share four returns, so h_t changes slowly by construction. This makes persistence a strong benchmark.
- **Large jumps in h_t .** The innovations of h_t are extremely heavy-tailed ($\nu \approx 1.8$). A single large return makes $\log RV$ jump when it enters the window and drop five days later, and these jumps cannot be predicted from past volatility.
- **Lagged regime indicator.** The crisis regime switches on only after a volatility spike has been observed. In the test period it is active on 18 days: 7–12 August 2024, after the market turbulence of early August [12]; 17 March 2025; 4–17 and 23–28 April 2025. The model reacts to crises but cannot anticipate them.

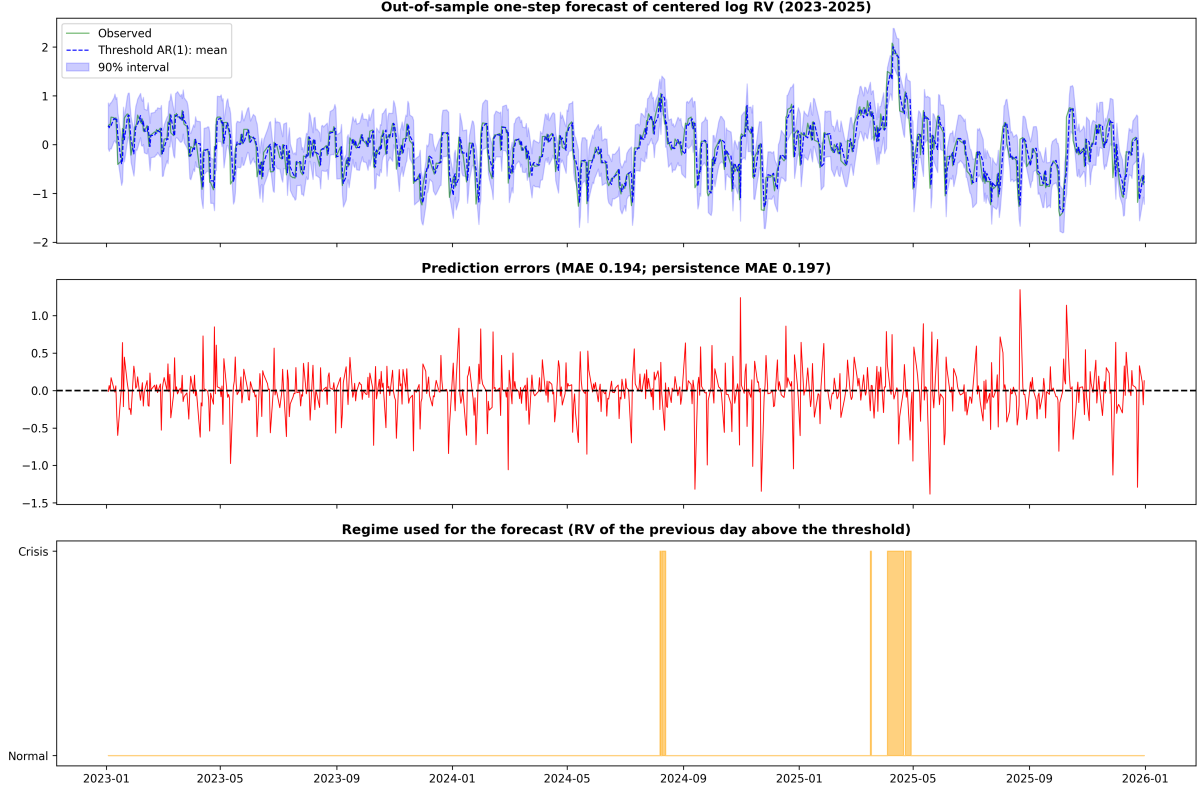


Figure 2: Out-of-sample one-step forecasts of centered log realized volatility with the threshold AR(1) (top), prediction errors (middle), and the regime used for each forecast (bottom).

5 Limitations and extensions

- **The SV models are not evaluated out of sample.** A comparison of their one-step-ahead forecasts, computed with a particle filter [3], would also assess their predictive performance.
- **A noisy volatility proxy.** Five-day realized volatility is noisy, and its windows overlap. Estimators based on daily high and low prices [13, 14] could reduce both problems.
- **No leverage effect.** Falling prices tend to raise future volatility more than rising prices do. The models treat both the same way; a correlation between ε_t and η_{t+1} could capture this asymmetry [15].

6 Conclusion

A Bayesian stochastic volatility model shows that the volatility of the S&P 500 varies strongly and persistently over time, and that this variation is the main source of the heavy tails of weekly returns: once volatility is allowed to vary, the return shocks are close to Normal. Discrete crisis regimes defined by a volatility threshold do not improve out-of-sample forecasts over a single-regime model, and both models are only marginally better than a naive persistence forecast. With this data, continuous time-varying volatility is the feature that matters.

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